IMPERIAL COLLEGE LONDON

EE4-10
EE9CS5-1
EE9SO26

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2012

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Friday, 18 May 2:30 pm

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Angeli

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PROBABILITY AND STOCHASTIC PROCESSES

1. The aim of the question is to analyse 3 different games.

a) In **Game A**, the player repeatedly flips a biased coin, coin a, wins \$1 if it comes up heads with probability $0 < p_a < 1/2$ and loses \$1 otherwise. $S_n^{(1)}$ is the wealth of the player in round n with $S_0^{(1)} = 0$, possibly negative if she lost money, after n rounds of the game. Game A is clearly a losing game, i.e. in the long run one is more likely to lose than to win.

i) Prove that the process $(S_n^{(1)})_{n\geq 0}$ is a Markov chain. [2]

ii) Let $W_n = \left(\frac{1-p_a}{p_a}\right)^{S_n^{(1)}}$.
Show that $\mathbf{E}\left(W_{n+1} \mid S_n^{(1)}\right) = W_n$. Determine $\mathbf{E}(W_n)$. [3]

iii) We consider the event E_1 that the wealth of the player reaches k before reaching $-k$ for a positive integer k. Show that

$$\mathbf{P}(E_1) = \frac{1 - \left(\frac{p_a}{1-p_a}\right)^k}{\left(\frac{1-p_a}{p_a}\right)^k - \left(\frac{p_a}{1-p_a}\right)^k}.$$

[4]

b) We now consider a more elaborate game, **Game B** involving 2 coins. The wealth of the player after n rounds is given by $S_n^{(2)}$, $S_0^{(2)} = 0$. If $S_n^{(2)}$ is a multiple of 3 then in round $n+1$ the player flips coin b that comes up heads with probability p_b and tails with probability $1 - p_b$, otherwise she flips coin c that comes up heads with probability p_c and tails with probability $1 - p_c$. The player wins \$1 if heads shows up in either configuration and loses \$1 otherwise.

i) Let z_i be the probability that the player ends up losing \$3 before winning \$3 known that she started with $S_0^{(2)} = i$, $i \in \{-3, -2, \ldots, 2, 3\}$. Show that

$$z_0 = \frac{(1-p_b)(1-p_c)^2}{(1-p_b)(1-p_c)^2 + p_b p_c^2}.$$

Hint: You do **not** *need to perform all the calculations. Instead only provide the main steps.* [4]

ii) For $p_b = 0.09$, $p_c = 0.74$, justify that Game B is a losing game. [2]

c) We now combine the two losing games A and B into a winning game. This paradox, known as *Parrondo's paradox*, appears to contradict the saying that two wrongs don't make a right!
In **Game C**, repeatedly perform the following bet. Flip a *fair* coin D. If heads shows up play Game A, otherwise play Game B.

i) Show that Game C has the same structure as Game B with probabilities p_b^* and p_c^* instead of p_b and p_c that you will express in terms of p_b and p_c. [3]

ii) For $p_a = 0.49$, $p_b = 0.09$ and $p_c = 0.74$, justify Game C is a winning game. Explain why this yields a paradox. [2]

2. We are interested in counting the number of entries equal to 1 that appear in a binary string. A straightforward approach is to count all the "1" entries one by one. It gives the exact answer but requires about $\log(n)$ bits of memory for a string with n entries. We propose instead to find an approximation $\tilde{H}$ such that $\mathbf{E}(\tilde{H})$ the average of $\tilde{H}$ is equal to H. Such procedure will instead require only about $\log(\log(n))$ bits of memory for a string with n entries. To this end we will use the following algorithm.

k := 0 *% Start the counter at* 0
For i = 1 to n *% Go through the n entries of the binary string*
Read x_i the *i*-th entry of the string
If $x_i = 1$ then with probability $1/2^k$ do k := k + 1
% Increment the current value of the counter k by 1 *with probability* $1/2^k$
Output $2^k - 1$

The input of the above procedure is the binary string x containing H entries equal to 1 and the output is a random variable $\tilde{H}$ that approximates H. We will analyse the distribution of the counter k in the above procedure. For $m = 0, \ldots, H$, let K_m be the value of the counter k after seeing m entries equal to one and let

$$p_{m,i} = \mathbf{P}(K_m = i), \quad i = 0, \ldots, m$$

For example $p_{0,0} = 1$ since the counter does not get incremented unless a 1 appears in the binary string, and $p_{m,0} = 0$ for $m = 1, \ldots, H$ since when a one is encountered for the first time then the counter is incremented with probability $1/2^0 = 1$.

a) We start by the following simple illustration. Consider the string 01011.

i) Determine $\mathbf{P}(K_1 = 1), \mathbf{P}(K_2 = 1)$. [1]

ii) Compute $\mathbf{P}(K_3 = 2 \mid K_2 = 1)$ and $\mathbf{P}(K_3 = 2 \mid K_2 = 2)$. [2]

iii) Determine $\mathbf{P}(K_3 = 2)$. [2]

b) We now look at the general case.

i) Determine $\mathbf{P}(K_m = i \mid K_{m-1} = i)$ and $\mathbf{P}(K_m = i \mid K_{m-1} = i-1)$. [2]

ii) Show that $p_{m,m} = \frac{1}{2^{m-1}} p_{m-1,m-1}$. [1]

iii) Using conditional probabilities, prove that, for $1 \leq i \leq m-1$

$$p_{m,i} = p_{m-1,i}\left(1 - \frac{1}{2^i}\right) + p_{m-1,i-1}\frac{1}{2^{i-1}}.$$

[3]

c) We introduce $C_m = 2^{K_m}$. In particular $\tilde{H} = C_H - 1$

i) Derive the values of $\mathbf{E}(C_0)$ and $\mathbf{E}(C_1)$. [1]

ii) Express $\mathbf{E}(C_m)$ in terms of of the $p_{m,i}$, $i = 1, \ldots, m$. [1]

iii) Using b)ii) and b)iii) show that

$$\mathbf{E}(C_m) = \sum_{i=1}^{m-1} 2^i p_{m-1,i} + \sum_{i=1}^{m-1} p_{m-1,i}.$$

[3]

iv) Show that $\mathbf{E}(C_m) = m + 1$. [3]

v) Conclude that $\mathbf{E}(\tilde{H}) = H$. [1]

3. The aim of this problem is to analyse a model of peer-to-peer networks. Users arrive to the system according to a Poisson process with rate λ and each user stays connected to the network for a duration that is exponentially distributed with parameter μ.

a) We first prove some useful properties about exponential distributions.

i) Determine $\mathbf{P}(X > x)$ the tail distribution of X exponentially distributed with parameter λ. [1]

ii) Show that if X_1, X_2 are independent exponentially distributed random variables with parameters λ_1, λ_2 respectively, then the random variable $\min(X_1, X_2)$ is exponentially distributed with parameter $\lambda_1 + \lambda_2$ and that

$$\mathbf{P}(\min(X_1, X_2) = X_1) = 1 - \mathbf{P}(\min(X_1, X_2) = X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

[2]

iii) Show that if $X_1, X_2, \ldots, X_n$ are independent exponentially distributed random variables with parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$ respectively, then the random variable $\min(X_1, \ldots, X_n)$ is exponentially distributed with parameter $\sum_{i=1}^n \lambda_i$ and that, for $k = 1, \ldots, n$,

$$\mathbf{P}(\min(X_1, \ldots, X_n) = X_k) = \frac{\lambda_k}{\sum_{i=1}^n \lambda_i}.$$

[2]

b) We can view the above model of peer-to-peer systems as a continuous-time Markovian queue with an unlimited number of servers. A customer starts being served the moment she joins the system and leaves when she is done. If there are $k \geq 0$ customers in the queue, then the next event can be either (i) a termination of service of one of the k current customers, each staying for an exponential duration with parameter μ or (ii) the arrival of a new customer that takes place after an exponential duration with parameter λ.

i) Determine the probability $p_{k,k+1}$ that the first among the above events is the arrival of a new customer. [2]

ii) Determine the probability $p_{k,k-1}$ that the first among the above events is the departure of a customer. [2]

iii) Determine the distribution of the duration of time before either event occurs. [2]

c) We will now derive the stationary distribution $\pi = (\pi_0, \pi_1, \pi_2, \ldots)$ of the Markov chain $(M(t), t \geq 0)$ representing the number of customers present in the system at time t.

i) Describe the rate matrix Q of $(M(t), t \geq 0)$. [2]

ii) Show that $\pi_0 \lambda = \pi_1 \mu$. [1]

iii) Show that $\pi_{k+1} = \frac{\lambda}{\mu(k+1)} \pi_k$ for $k \geq 1$. [3]

iv) Conclude that $\pi_k = e^{-\lambda/\mu} \frac{1}{k!} \left(\frac{\lambda}{\mu}\right)^k$. [3]

SOLUTIONS

PROBABILITY & STOCHASTIC PROCESSES 2011/2012

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Q1

Q1 1/2

1)

a) Game A

i) $X_k^{(A)}$ outcome of kth round

$$\mathbb{P}(X_k^{(A)} = A) = 1 - \mathbb{P}(X_k^{(A)} = -A) = p_A.$$

$$S_n^{(A)} = S_{n-1}^{(A)} + X_n^{(A)}$$

So $(S_n^{(A)})_{n\geq 0}$ is clearly a Markov chain. [2]

ii) $$\mathbb{E}(W_{n+1} \mid S_n^{(A)}) = \mathbb{E}\left(\left(\frac{1-p_A}{p_A}\right)^{S_{n+1}^{(A)}} \mid S_n^{(A)}\right).$$

$$= \mathbb{E}\left(\left(\frac{1-p_A}{p_A}\right)^{S_n^{(A)} + X_{n+1}^{(A)}} \mid S_n^{(A)}\right)$$

$$= \left(\frac{1-p_A}{p_A}\right)^{S_n^{(A)}} \left(\underbrace{p_A \frac{1-p_A}{p_A} + (1-p_A)\frac{p_A}{1-p_A}}_{=2}\right)$$

$$= W_n.$$

$\mathbb{E}(W_n) = \mathbb{E}(W_{n-1}) = \mathbb{E}(W_0) = 1$; $W_0 = \left(\frac{1-p_A}{p_A}\right)^0 = 1$ [3]

iii). At time T when either k or $-k$ is reached $\mathbb{P}(E_A) = \mathbb{P}(S_T^{(A)} = k)$.

$$= 1 - (\mathbb{P}(S_T^{(A)} = -k)) = 1 - \mathbb{P}(\bar{E}_A)$$

$$\mathbb{E}(W_T) = 1$$

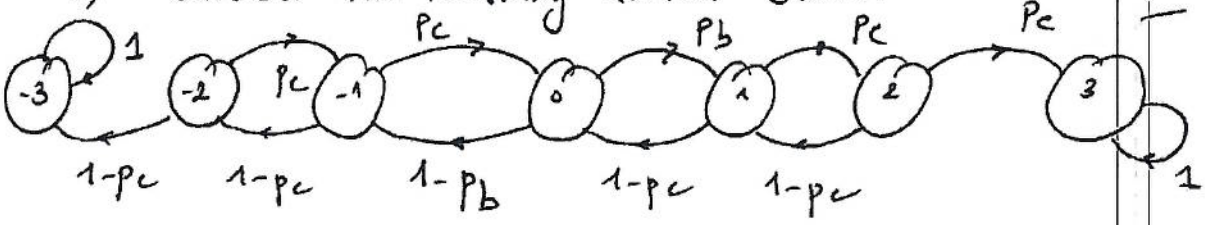
$$\mathbb{E}(W_T) = \mathbb{P}(E_A)\left(\frac{1-p_A}{p_A}\right)^k + (1-\mathbb{P}(E_A))\left(\frac{p_A}{1-p_A}\right)^k.$$

$$\mathbb{P}(E_A)\left[\left(\frac{1-p_A}{p_A}\right)^k - \left(\frac{p_A}{1-p_A}\right)^k\right] = 1 - \left(\frac{p_A}{1-p_A}\right)^k$$ [4]

$$\Rightarrow \mathbb{P}(E_A) = \frac{1 - \left(p_A/1-p_A\right)^k}{\left(\frac{1-p_A}{p_A}\right)^k - \left(\frac{p_A}{1-p_A}\right)^k}.$$

b) Game B:

i) Consider the following Markov chain.



$z_{-2} = (1-p_c)\, z_{-3} + p_c\, z_{-1}$; $z_{-3} = 1$

$z_{-1} = (1-p_c)\, z_{-2} + p_c\, z_0$

$z_0 = p_b\, z_1 + (1-p_b)\, z_{-1}$

$z_1 = (1-p_c)\, z_0 + p_c\, z_2$

$z_2 = (1-p_c)\, z_3 + p_c\, z_1$; $z_3 = 0$.

~~$(z_{-1} = (1-p_c)^2 + (1-p_c)\, p_c\, z_{-1} + p_c\, z_0.)$~~

Solving the above system we obtain.

$$z_0 = \frac{(1-p_b)(1-p_c)^2}{(1-p_b)(1-p_c)^2 + p_b p_c^2}.$$

[4]

ii) $z_0 = 0.55$, showing that she is more likely to lose than win playing this game over the long run. It is a losing game. [2]

c) i) In Game C, if winnings are multiple of 3, then playing either Game A or Game B; each w.p. 1/2; the probability of winning is $p_b^* = \frac{1}{2} p_a + \frac{1}{2} p_b$. Otherwise the prob. of winning is $p_c^* = \frac{1}{2} p_a + \frac{1}{2} p_c$ [3]

ii) $z_0^* = \dfrac{(1-p_b^*)(1-p_c^*)^2}{(1-p_b^*)(1-p_c^*)^2 + p_b^* p_c^{*2}} =$ ~~0.388~~ 0.49

$p_b^* = \frac{0.49 + 0.09}{2} = 0.29$; $p_c^* = \frac{0.49 + 0.74}{2} = 0.615$

More likely to win than to lose ⇒ Winning game. [2]

Q2 1/2

Q2 :

$k = 0$

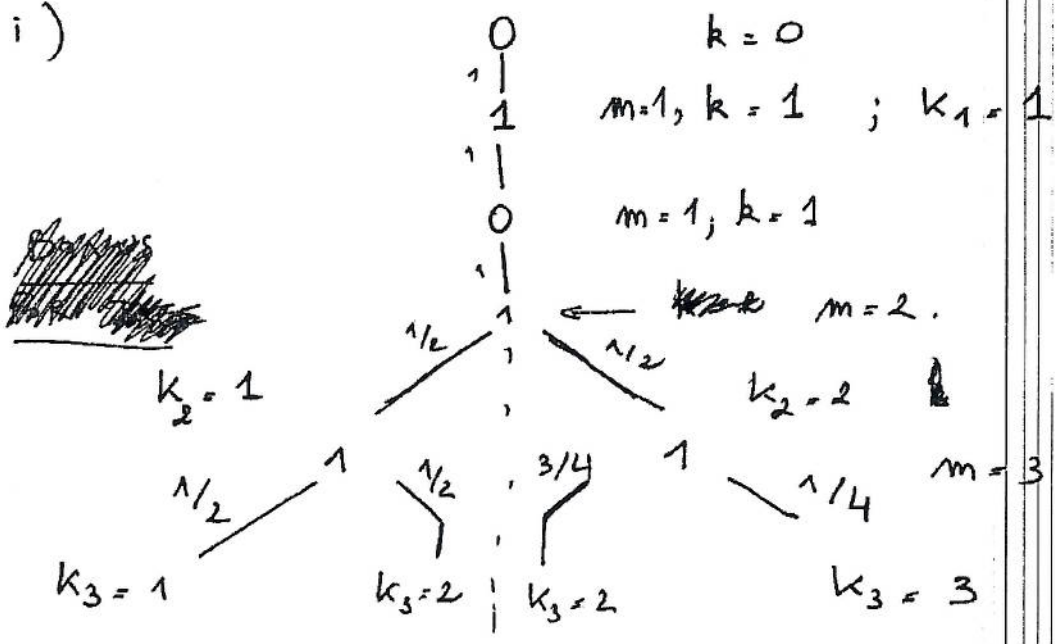
$i = 1$ to n.

$x_i = 1$ w.p. $1/2^k$ $\quad k \leftarrow k+1$.

k_m = value of counter after m 1s.

$p_{m,i} = \mathbb{P}(k_m = i)$.

a) i)



$\mathbb{P}(k_1 = 1) = 1$; $\mathbb{P}(k_2 = 1) = 1/2$. [1]

ii) $\mathbb{P}(k_3 = 2 \mid k_2 = 1) = 1/2$; $\mathbb{P}(k_3 = 2 \mid k_2 = 2) = 3/4$. [2]

iii) $\mathbb{P}(k_3 = 2) = \mathbb{P}(k_3 = 2 \mid k_2 = 1)\,\mathbb{P}(k_2 = 1) + \mathbb{P}(k_3 = 2 \mid k_2 = 2)\,\mathbb{P}(k_2$

$= 1/2 \cdot 1/2 + 3/4 \times 1/2 = 5/8$ [2]

b) i) $\mathbb{P}(k_m = i \mid k_{m-1} = i) = \dfrac{2^i - 1}{2^i} = 1 - \dfrac{1}{2^i}$

$\mathbb{P}(k_m = i \mid k_{m-1} = i-1) = \dfrac{1}{2^{i-1}}$. [2]

ii) $\mathbb{P}(k_m = m) = \mathbb{P}(k_m = m \mid k_{m-1} = m-1)\, \mathbb{P}(k_{m-1} = m-1 \mid k_{m-1} = m-1)$

$= \frac{1}{2^{m-1}}\, p_{m-1,m-1}$ [1] Q2 2/2

iii) $1 \le i \le m-1$

$\mathbb{P}(k_m = i) = \mathbb{P}(k_m = i \mid k_{m-1} = i)\, \mathbb{P}(k_{m-1} = i) + \mathbb{P}(k_m = i \mid k_{m-1} = i-1)\, \mathbb{P}(k_{m-1} = i-1)$

$= \left(1 - \frac{1}{2^i}\right) p_{m-1,i} + \frac{1}{2^{i-1}}\, p_{m-1,i-1}$. [3]

c) i) $\mathbb{E}(C_0) = \mathbb{E}(2^{k_0}) = 1 \qquad k_0 = 0$

$\mathbb{E}(C_1) = \mathbb{E}(2^{k_1}) = 2 \qquad k_1 = 1$ [1]

ii) $\mathbb{E}(C_m) = \mathbb{E}(2^{k_m}) = \sum_{i=0}^{m} 2^i p_{m,i} = \sum_{i=1}^{m} 2^i p_{m,i} \qquad p_{m,0} = 0$ [1]

iii) & iv)

$$\mathbb{E}(C_m) = \sum_{i=1}^{m-1} 2^i \left(\left(1 - \frac{1}{2^i}\right) p_{m-1,i} + \frac{1}{2^{i-1}}\, p_{m-1,i-1}\right) + 2^m \frac{1}{2^{m-1}}\, p_{m-1,m-1}$$

iii) [3] $= \underbrace{\sum_{i=1}^{m-1} 2^i\, p_{m-1,i}}_{\mathbb{E}(C_{m-1})} - \underbrace{\sum_{i=1}^{m-1} p_{m-1,i}}_{1} + \underbrace{\sum_{i=1}^{m-1} 2\, p_{m-1,i-1} + 2\, p_{m-1,m-1}}_{2}$

iv) [3] $= \mathbb{E}(C_{m-1}) + 1 = m+1$ donc $\mathbb{E}(G) = +\infty$

v) $\mathbb{E}(\tilde{C}) = \mathbb{E}(C_{H-1}) = \mathbb{E}(2^H - 1) = H+1-1 = +\infty$ [1]

Q3 1/2

a) i) $\mathbb{P}(X > x) = \int_x^{-\infty} \lambda e^{-\lambda t}\, dt = e^{-\lambda x}$ $x \geq 0$.

[1]

ii) $\mathbb{P}(\min(X_1, X_2) > x) = \mathbb{P}(X_1 > x)\, \mathbb{P}(X_2 > x)$

$= e^{-(\lambda_1 + \lambda_2)x}$.

$\min(X_1, X_2) \sim Exp(\lambda_1 + \lambda_2)$. [2]

$\mathbb{P}(\min(X_1, X_2) = X_1) = \mathbb{P}(X_1 < X_2)$

$= \int_0^{\infty} \lambda_1 e^{-\lambda_1 x} \int_x^{\infty} \mathbb{P}(x < X_2)\, dy$

$= \lambda_1 \int_0^{+\infty} e^{-\lambda_1 x} e^{-\lambda_2 x}\, dx$

$= \frac{\lambda_1}{\lambda_1 + \lambda_2}$.

iii) By induction. [2]

b) i) $p_{k,k+1} = \mathbb{P}(Exp(\lambda) < \underbrace{Exp(k\mu)}_{\min_k Exp(\mu)})$

$= \frac{\lambda}{\lambda + k\mu}$. [2]

ii) $p_{k,k-1} = \frac{k\mu}{\lambda + k\mu}$. [2]

iii) $Exp(\lambda + k\mu) = \min\{Exp(\lambda), Exp(k\mu)\}$.

[2]

c)

i) $Q = \begin{bmatrix} -\lambda & \lambda & 0 & \cdots & \\ \mu & -(\lambda+\mu) & \mu & 0 & \cdots \\ 0 & 2\mu & -(\lambda+2\mu) & 2\mu & 0 \cdots \\ & \ddots & \ddots & \ddots & \\ & & k\mu & -(\lambda+k\mu) & k\mu \end{bmatrix}$ 2/3 [2]

ii) iii) $\pi Q = 0$.

$-\lambda \pi_0 + \mu \pi_1 = 0 \Rightarrow \pi_0 \lambda = \mu \pi_1$. [1]

$\pi_{k+1}(k+1)\mu - \pi_k(\lambda + k\mu) + \lambda \pi_{k-1} = 0$.

$\Rightarrow \pi_k k\mu = \pi_{k-1}\lambda$.

$\Rightarrow \pi_{k+1} = \dfrac{\lambda}{\mu(k+1)} \pi_k \quad k \geq 1$. [3]

iv) $\pi_k = \pi_0 \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!} \pi_0$.

$1 = \sum \pi_k = \pi_0 \sum \left(\lambda/\mu\right)^k \frac{1}{k!} = \pi_0 e^{\lambda/\mu}$.

$\pi_0 = e^{-\lambda/\mu}$

$\Rightarrow \pi_k = e^{-\lambda/\mu} \frac{1}{k!} \left(\lambda/\mu\right)^k$.

Stat dist is Poisson (λ/μ). [3]